Nonlinear Dyn (2012) 69:1621–1630
DOI 10.1007/s11071-012-0373-7

ORIGINAL PAPER

Darboux transformation for an integrable generalization of the nonlinear Schrödinger equation

Xianguo Geng · Yanyan Lv

Received: 17 December 2011 / Accepted: 16 February 2012 / Published online: 10 March 2012
© Springer Science+Business Media B.V. 2012

Abstract A Darboux transformation for an integrable generalization of the coupled nonlinear Schrödinger equation is derived with the help of the gauge transformation between the Lax pair. As a reduction, a Darboux transformation for an integrable generalization of the nonlinear Schrödinger equation is obtained, from which some new solutions for the integrable generalization of the nonlinear Schrödinger equation are given.

Keywords An integrable generalization of the nonlinear Schrodinger equation · Darboux transformation · Exact solutions

1 Introduction

In [1], Fokas proposed an integrable generalization of the nonlinear Schrödinger (NLS) equation

$$iu_t - \nu u_{tx} + \gamma u_{xx} + \sigma|u|^2(u + i\nu u_x) = 0, \quad \sigma = \pm 1, \tag{1}$$

by using bi-Hamiltonian methods, where γ and ν are nonzero real parameters and $u(x,t)$ is a complex-valued function. Equation (1) reduces to the NLS equation when $\nu = 0$, and it arises as a model for nonlinear pulse propagation in monomode optical fibers and is the first negative member of the integrable hierarchy associated with the derivative NLS equation [2]. In [3], Lenells and Fokas used the bi-Hamiltonian structure to write down the first few conservation laws of (1) and derive its Lax pair, by which they solve the initial value problem and analyze solitons. The initial-boundary value problem for (1) on the half-line was studied in [4]. In particular, the so-called linearizable boundary conditions were considered. Furthermore, a particular solution was used to verify explicitly all the steps needed for the solution of a well-posed problem. Recently, Lenells implemented the dressing method for (1) and derived explicit formulas for the N-soliton solutions [5].

The aim of the present paper is to study the integrable generalization of the NLS equation with the aid of the Darboux transformation (DT). It is known that the DT is a powerful tool which generates new solutions of an integrable equation from an already known seed solution [6–10]. The DT approach possesses a lot of advantages compared with other methods. For example, if the obtained solutions are taken as the new starting point, one can make the DT once again and engender another set of new solutions. This process can be done continually and will usually yield a series of new explicit solutions. Here, we first construct a DT for an integrable generalization of the coupled NLS equation with the aid of a gauge transformation between the corresponding matrix spectral problems. As

X. Geng (✉) · Y. Lv
Department of Mathematics, Zhengzhou University, 100 Kexue Road, Zhengzhou, Henan 450001, People's Republic of China
e-mail: xggeng@zzu.edu.cn

a reduction, a DT for the integrable generalization of the NLS equation is obtained. A systematic algebraic procedure is given in detail to solve the integrable generalization of the NLS equation, from which some new explicit solutions are derived.

The present paper is organized as follows. In Sect. 2, we shall construct a DT for the integrable generalization of the coupled NLS equation resorting to the gauge transformation between matrix spectral problems. In Sect. 3, a DT for the integrable generalization of the NLS equation is discussed through the reduction technique. Furthermore, two classes of new explicit solutions for the integrable generalization of the NLS equation are derived by the application of its DT.

2 Darboux transformation

In this section, we shall construct a DT for an integrable generalization of the coupled NLS equation

$$\begin{aligned} &u_{xt}+\alpha\beta^2u-2i\alpha\beta u_x-\alpha u_{xx}+i\alpha\beta^2uvu_x=0,\\ &v_{xt}+\alpha\beta^2v+2i\alpha\beta v_x-\alpha v_{xx}-i\alpha\beta^2uvv_x=0, \end{aligned} \tag{2}$$

which comes from the condition of the compatibility for two matrix spectral problems [3]:

$$\psi_x=M\psi,\quad M=\begin{pmatrix}-i\zeta^2 & \zeta u_x\\ \zeta v_x & i\zeta^2\end{pmatrix}, \tag{3}$$

$$\begin{aligned} &\psi_t=N\psi,\\ &N=\begin{pmatrix}-\frac{i\alpha\beta^2}{2}uv-i\eta^2 & \alpha\zeta u_x+\frac{i\alpha\beta^2}{2\zeta}u\\ \alpha\zeta v_x-\frac{i\alpha\beta^2}{2\zeta}v & \frac{i\alpha\beta^2}{2}uv+i\eta^2\end{pmatrix}, \end{aligned} \tag{4}$$

where u and v are two potentials with independent variables x and t, ζ is a constant spectral parameter, and

$$\eta=\sqrt{\alpha}\left(\zeta-\frac{\beta}{2\zeta}\right). \tag{5}$$

Under a certain transformation (see Sect. 3 below), (2) is reduced the integrable generalization of the NLS equation (1), which has important applications in physics.

In what follows, we shall derive a DT for the integrable generalization of the coupled NLS equation (2). Let $\phi=(\phi_1,\phi_2)^T$, $\psi=(\psi_1,\psi_2)^T$ be two basic solution of (3). We use (ϕ,ψ) to define the linear algebraic system

$$\begin{aligned} &\sum_{k=1}^{N}\left(A_k\zeta_j^{2k}+\alpha_jB_k\zeta_j^{2k-1}\right)=-1,\quad j=1,\ldots,2N,\\ &\sum_{k=1}^{N}\left(C_k\zeta_j^{2k-1}+\alpha_jD_k\zeta_j^{2k}\right)=-\alpha_j,\quad j=1,\ldots,2N, \end{aligned} \tag{6}$$

with

$$\alpha_j=\frac{\phi_2(\zeta_j)-\gamma_j\psi_2(\zeta_j)}{\phi_1(\zeta_j)-\gamma_j\psi_1(\zeta_j)},\quad j=1,\ldots,2N, \tag{7}$$

where parameters ζ_j and γ_j ($\zeta_k\neq\zeta_j,\gamma_k\neq\gamma_j,k\neq j$) are suitably chosen such that the determinants of the coefficients for (6) are nonzero. Hence, A_k,B_k,C_k, and D_k ($k=1,\ldots,N$) are uniquely determined by (6). Now we introduce a gauge transformation of the spectral problem (3)

$$\tilde{\psi}=T\psi,\quad T=\begin{pmatrix}A & B\\ C & D\end{pmatrix}, \tag{8}$$

where

$$\begin{aligned} &A=1+\sum_{k=1}^{N}A_k\zeta^{2k},\qquad B=\sum_{k=1}^{N}B_k\zeta^{2k-1},\\ &C=\sum_{k=1}^{N}C_k\zeta^{2k-1},\qquad D=1+\sum_{k=1}^{N}D_k\zeta^{2k}. \end{aligned}$$

From (6), we have

$$\begin{aligned} &\left(1+\sum_{k=1}^{N}A_k\zeta_j^{2k}\right)\left(1+\sum_{k=1}^{N}D_k\zeta_j^{2k}\right)\\ &\quad=\sum_{k=1}^{N}\sum_{l=1}^{N}B_kC_l\zeta_j^{2k+2l-2}, \end{aligned}$$

that is

$$\begin{aligned} &1+\sum_{k=1}^{N}(A_k+D_k)\zeta_j^{2k}+\sum_{k=1}^{N}\sum_{l=1}^{N}\big(A_kD_l\zeta_j^{2k+2l}\\ &\quad-B_kC_l\zeta_j^{2k+2l-2}\big)=0. \end{aligned} \tag{9}$$

Equation (9) implies that ζ_j^2 ($j=1,\ldots,2N$) are $2N$ roots of the $(4N)$th-order polynomial $\det T$. Therefore, we have

$$\det T=A_ND_N\prod_{j=1}^{2N}\left(\zeta^2-\zeta_j^2\right). \tag{10}$$

Under the gauge transformation (8), the spectral problem (3) is transformed into the following spectral problem of $\tilde{\psi}$ when $\zeta\neq\zeta_j$ ($j=1,\ldots,2N$):

$$\tilde{\psi}_x = \tilde{M}\tilde{\psi}, \tag{11}$$

where

$$\tilde{M} = (T_x + TM)T^{-1}. \tag{12}$$

It turns out that $\zeta = \zeta_j$ $(j = 1, \ldots, 2N)$ are removable isolated singularities of $\tilde{M}$ (see (13) below). Thus, we can define $\tilde{M}$ for all ζ by analytic continuation.

Theorem 1 *The matrix $\tilde{M}$ determined by (12) has the same form as M:*

$$\tilde{M} = \begin{pmatrix} -i\zeta^2 & \zeta\tilde{u}_x \\ \zeta\tilde{v}_x & i\zeta^2 \end{pmatrix}, \tag{13}$$

where the transformation formulae from the old potentials u and v into new ones are given by

$$\tilde{u} = u + B_1, \qquad \tilde{v} = v + C_1. \tag{14}$$

The transformation determined by (8) and (14): $(\psi; u, v) \to (\tilde{\psi}; \tilde{u}, \tilde{v})$ is usually called a DT of the spectral problem (4).

Proof Let $T^{-1} = T^*/\det T$ and

$$(T_x + TM)T^* = \begin{pmatrix} f_{11}(\zeta) & f_{12}(\zeta) \\ f_{21}(\zeta) & f_{22}(\zeta) \end{pmatrix}. \tag{15}$$

A direct calculation shows that $f_{11}(\zeta)$, $f_{22}(\zeta)$, $\zeta f_{12}(\zeta)$, $\zeta f_{21}(\zeta)$ are $(2N+1)$th-order polynomials in ζ^2. By using (3), (6), and (7), we have

$$\begin{aligned} &\alpha_{jx} = \left(v_x - \alpha_j^2 u_x\right)\zeta_j + 2i\zeta_j^2\alpha_j, \\ &\quad j = 1, \ldots, 2N, \end{aligned} \tag{16}$$

$$A(\zeta_j) = -\alpha_j B(\zeta_j), \quad C(\zeta_j) = -\alpha_j D(\zeta_j). \tag{17}$$

Resorting to (16) and (17), it can be verified that ζ_j $(j = 1, \ldots, 2N)$ are roots of $f_{kl}(k, l = 1, 2)$ in (15). Therefore, we have from (10) that

$$(T_x + TM)T^* = (\det T)P(\zeta), \tag{18}$$

with

$$P(\zeta) = \begin{pmatrix} p_{11}^{(2)}\zeta^2 + p_{11}^{(0)} & p_{12}^{(1)}\zeta \\ p_{21}^{(1)}\zeta & p_{22}^{(2)}\zeta^2 + p_{22}^{(0)} \end{pmatrix},$$

where $p_{kl}^{(j)}$ are independent of ζ. Equation (18) can be written as

$$T_x + TM = P(\zeta)T. \tag{19}$$

Equating the coefficients of ζ^{2N+2}, ζ, and ζ^0 in (19), we obtain

$$p_{11}^{(2)} = -i, \qquad p_{22}^{(2)} = i, \qquad p_{11}^{(0)} = p_{22}^{(0)} = 0, \tag{20}$$

$$\begin{aligned} &p_{12}^{(1)} = -p_{11}^{(0)}B_1 + B_{1x} + u_x, \\ &p_{21}^{(1)} = -p_{22}^{(0)}C_1 + C_{1x} + v_x. \end{aligned} \tag{21}$$

Substituting (14) and (20) into (21) yields

$$p_{12}^{(1)} = \tilde{u}_x, \qquad p_{21}^{(1)} = \tilde{v}_x. \tag{22}$$

The proof is thus completed. □

Remark 1 Equating the coefficients of ζ^{2N+1} in (19), we have

$$\begin{aligned} &A_N u_x + iB_N = p_{11}^{(2)}B_N + p_{12}^{(1)}D_N, \\ &-iC_N + v_x D_N = p_{21}^{(1)}A_N + p_{22}^{(2)}C_N. \end{aligned} \tag{23}$$

By virtue of (20) and (22), we obtain

$$\tilde{u}_x = \frac{A_N u_x + 2iB_N}{D_N}, \qquad \tilde{v}_x = \frac{D_N v_x - 2iC_N}{A_N}. \tag{24}$$

Assume that the two solutions ϕ and ψ of (3) satisfy (4) as well. Differentiating $\tilde{\psi} = T\psi$ with respect to t yields

$$\tilde{\psi}_t = \tilde{N}\tilde{\psi}, \qquad \tilde{N} = (T_t + TN)T^{-1}. \tag{25}$$

Then we can prove the following assertion.

Theorem 2 *The matrix $\tilde{N}$ defined by the second expression of (25) has the same form as N, in which the old potentials u and v are mapped into $\tilde{u}$ and $\tilde{v}$ according to the same DT (14).*

Proof Let $T^{-1} = T^*/\det T$ and

$$(T_t + TN)T^* = \begin{pmatrix} g_{11}(\zeta) & g_{12}(\zeta) \\ g_{21}(\zeta) & g_{22}(\zeta) \end{pmatrix}. \tag{26}$$

A direct calculation shows that $\zeta^2 g_{11}(\zeta)$ and $\zeta^2 g_{22}(\zeta)$ are $(2N+2)$th-order polynomials in ζ^2, $\zeta g_{12}(\zeta)$, and $\zeta g_{21}(\zeta)$ are $(2N+1)$th-order polynomials in ζ^2. By virtue of (4), (7), and (17), we have

$$\begin{aligned} \alpha_{jt} &= \alpha\zeta_j v_x - \frac{i\alpha\beta^2}{2\zeta_j}v + 2\alpha_j\left(\frac{i\alpha\beta^2}{2}uv + i\eta_j^2\right) \\ &\quad -\left(\alpha\zeta_j u_x + \frac{i\alpha\beta^2}{2\zeta_j}u\right)\alpha_j^2, \\ A_t(\zeta_j) &= -\alpha_{jt}B(\zeta_j) - \alpha_j B_t(\zeta_j), \\ C_t(\zeta_j) &= -\alpha_{jt}D(\zeta_j) - \alpha_j D_t(\zeta_j), \end{aligned} \tag{27}$$

where $\eta_j = \sqrt{\alpha}(\zeta_j - \frac{\beta}{2\zeta_j})$. It can be easily verified by (16), (17), and (27) that ζ_j $(j = 1, \ldots, 2N)$ are roots of $g_{kl}(\zeta)$, $k, j = 1, 2$. Therefore, we arrive at

$$(T_t + TN)T^* = (\det T)Q(\zeta), \tag{28}$$

with

$$Q(\zeta)=\begin{pmatrix} q_{11}^{(2)}\zeta^2+q_{11}^{(0)}+q_{11}^{(-2)}\zeta^{-2} & q_{12}^{(1)}\zeta+q_{12}^{(-1)}\zeta^{-1} \\ q_{21}^{(1)}\zeta+q_{21}^{(-1)}\zeta^{-1} & q_{22}^{(2)}\zeta^2+q_{22}^{(0)}+q_{22}^{(-2)}\zeta^{-2} \end{pmatrix},$$

where $q_{kl}^{(j)}$ are independent of ζ. Equation (28) can be written as

$$T_t+TN=Q(\zeta)T. \tag{29}$$

By comparing the coefficients of $\zeta^{2N+2},\zeta^{-2},\zeta^{-1}$, and ζ^0 in (29), we get

$$\begin{aligned} &q_{11}^{(2)}=-i\alpha, \qquad q_{22}^{(2)}=i\alpha, \\ &q_{11}^{(-2)}=-\frac{i\alpha\beta^2}{4}, \qquad q_{22}^{(-2)}=\frac{i\alpha\beta^2}{4}, \end{aligned} \tag{30}$$

$$\begin{aligned} &q_{12}^{(-1)}=\frac{i\alpha\beta^2}{2}(u+B_1), \\ &q_{21}^{(-1)}=-\frac{i\alpha\beta^2}{2}(v+C_1), \\ &q_{11}^{(0)}=i\alpha\beta-\frac{i\alpha\beta^2}{2}v(u+B_1)-q_{12}^{(-1)}C_1, \\ &q_{22}^{(0)}=-i\alpha\beta+\frac{i\alpha\beta^2}{2}u(v+C_1)-q_{21}^{(-1)}B_1. \end{aligned} \tag{31}$$

By using (14), we obtain

$$\begin{aligned} &q_{12}^{(-1)}=\frac{i\alpha\beta^2}{2}\tilde{u}, \qquad q_{21}^{(-1)}=-\frac{i\alpha\beta^2}{2}\tilde{v}, \\ &q_{11}^{(0)}=i\alpha\beta-\frac{i\alpha\beta^2}{2}\tilde{u}\tilde{v}, \qquad q_{22}^{(0)}=-i\alpha\beta+\frac{i\alpha\beta^2}{2}\tilde{u}\tilde{v}. \end{aligned} \tag{32}$$

Equating the coefficients of ζ^{2N+1} in (29), we have

$$\begin{aligned} &q_{12}^{(1)}=\alpha\frac{A_Nu_x+2iB_N}{D_N}, \\ &q_{21}^{(1)}=\alpha\frac{D_Nv_x-2iC_N}{A_N}=\alpha\tilde{v}_x, \end{aligned}$$

which together with (24) implies

$$q_{12}^{(1)}=\alpha\tilde{u}_x, \qquad q_{21}^{(1)}=\alpha\tilde{v}_x. \tag{33}$$

The proof is thus completed. □

According to Theorems 1 and 2, the transformation (8) and (14) converts the Lax pair (3) and (4) into another Lax pair of the same type (11) and (25). Therefore, both of the Lax pairs lead to the same integrable generalization of the coupled NLS equation (2). The transformation (14) is also called a DT for the integrable generalization of the coupled NLS equation (2). We get immediately the following fact.

Theorem 3 *Every solution (u,v) for the integrable generalization of the coupled NLS equation (2) is mapped into a new solution $(\tilde{u},\tilde{v})$ for the integrable generalization of the coupled NLS equation (2) under the DT (14), where B_1 and C_1 are given by the linear algebraic system (6).*

3 The reduction of Darboux transformation and application

In this section, we will use the reduction technique to derive a DT for the integrable generalization of the NLS equation. Let $\alpha=\frac{\gamma}{v}>0$ and $\beta=\frac{1}{v}$. Then the transformation, $u\to\beta\sqrt{\alpha}\,e^{i\beta x}u$, $\sigma\to-\sigma$, converts (1) into [5]

$$\begin{aligned} &u_{tx}+\alpha\beta^2u-2i\alpha\beta u_x-\alpha u_{xx}+\sigma i\alpha\beta^2|u|^2u_x=0, \\ &\sigma=\pm1, \end{aligned} \tag{34}$$

which is equivalent to (1) because the above transformation is invertible. Without loss of generality, in the rest of the paper we will concentrate on the integrable generalization of the NLS equation (34). It is easy to see that (2) reduces to (34) as well when $v=\sigma u^*$. Therefore, we always assume that $v=\sigma u^*$ as follows. In this case, we choose two solutions of (3) and (4):

$$\begin{aligned} &\phi(\zeta)=\big(\phi_1(\zeta),\phi_2(\zeta)\big)^T, \\ &\psi(\zeta)=\big(\sigma\phi_2^*\big(\zeta^*\big),\phi_1^*\big(\zeta^*\big)\big)^T, \end{aligned} \tag{35}$$

and parameters

$$\zeta_{2j}=\zeta_{2j-1}^*, \qquad \gamma_{2j}=\sigma\gamma_{2j-1}^{*-1}, \quad j=1,\ldots,N. \tag{36}$$

By using (6) and (7), we obtain $\alpha_{2j}^{-1}=\sigma\alpha_{2j-1}^*$, $D_k=A_k^*$, $C_k=\sigma B_k^*$, $(j=1,\ldots,N;k=1,\ldots,N)$. Here, α_{2j-1} is given by

$$\alpha_{2j-1}=\frac{\phi_2(\zeta_{2j-1})-\gamma_{2j-1}\phi_1^*(\zeta_{2j-1}^*)}{\phi_1(\zeta_{2j-1})-\sigma\gamma_{2j-1}\phi_2^*(\zeta_{2j-1}^*)}, \tag{37}$$

functions A_k and B_k are determined by the following linear algebraic system

$$\begin{aligned}&\sum_{k=1}^{N}\left(A_k\zeta_{2j-1}^{2k}+\alpha_{2j-1}B_k\zeta_{2j-1}^{2k-1}\right)=-1,\\&\sum_{k=1}^{N}\left(\alpha_{2j-1}^{*}A_k\zeta_{2j-1}^{*2k}+\sigma B_k\zeta_{2j-1}^{*2k-1}\right)=-\alpha_{2j-1}^{*},\\&\quad j=1,\ldots,N.\end{aligned}\tag{38}$$

According to (14) and $v=\sigma u^*$, we arrive at $\tilde{v}=v+C_1=\sigma u^*+\sigma B_1^*=\sigma\tilde{u}^*$, which implies that $\tilde{v}=\sigma\tilde{u}^*$. Therefore, we prove easily the following assertion.

Theorem 4 *Every solution u for the integrable generalization of the NLS equation (34) is mapped into a new solution $\tilde{u}$ under the DT*

$$\tilde{u}=u+B_1,\tag{39}$$

where B_1 is determined by (38).

In the following, we shall apply the DT to give explicit solutions for the integrable generalization of the NLS equation (34). Substituting the trivial solution $u=0$ ($v=\sigma u^*=0$) of (34) into (3) and (4), their two basic solutions $\phi(\zeta)$ and $\psi(\zeta)$ are chosen as

$$\begin{aligned}&\phi(\zeta)=\begin{pmatrix}\exp(-i\zeta^2x-i\eta^2t)\\0\end{pmatrix},\\&\psi(\zeta)=\begin{pmatrix}0\\\exp(i\zeta^2x+i\eta^2t)\end{pmatrix},\end{aligned}\tag{40}$$

in the view of (35). Then (37) and (38) can be written as

$$\alpha_{2j-1}=-\gamma_{2j-1}\exp\left(2i\zeta_{2j-1}^2x+2i\eta_{2j-1}^2t\right),\tag{41}$$

$$\begin{aligned}&\sum_{k=1}^{N}\left(A_k\zeta_{2j-1}^{2k}+\alpha_{2j-1}B_k\zeta_{2j-1}^{2k-1}\right)=-1,\\&\sum_{k=1}^{N}\left(\alpha_{2j-1}^{*}A_k\zeta_{2j-1}^{*2k}+\sigma B_k\zeta_{2j-1}^{*2k-1}\right)=-\alpha_{2j-1}^{*},\\&\quad j=1,\ldots,N.\end{aligned}\tag{42}$$

Using the DT, we obtain exact solutions for the integrable generalization of the NLS equation (34):

$$\tilde{u}=u+B_1=B_1,\tag{43}$$

where B_1 is determined by (42). For simplicity, we shall discuss the two special cases $N=1$ and $N=2$.

(I) For $N=1$, a direct calculation gives from (42) that

$$B_1=\frac{\alpha_1^*(\zeta_1^{*2}-\zeta_1^2)}{\zeta_1\zeta_1^*(\sigma\zeta_1-\alpha_1\alpha_1^*\zeta_1^*)}.$$

According to the Darboux transformation (39), we get a one-soliton solution for the integrable generalization of the NLS equation (34)

$$\tilde{u}=\frac{\alpha_1^*(\zeta_1^{*2}-\zeta_1^2)}{\zeta_1\zeta_1^*(\sigma\zeta_1-\alpha_1\alpha_1^*\zeta_1^*)},\tag{44}$$

with $\alpha_1=-\gamma_1\exp(2i\zeta_1^2x+2i\eta_1^2t)$, $\eta_1=\sqrt{\alpha}(\zeta_1-\frac{\beta}{2\zeta_1})$. Plots of this solution are given in Fig. 1 if the parameters are chosen as $\alpha=1,\beta=-1,\sigma=-1,\zeta_1=0.6+0.2i,\gamma_1=0.5+0.8i$ (see Fig. 1).

(II) For $N=2$, we obtain from linear algebraic system (42) that

$$\tilde{u}=B_1=\frac{\Delta_1}{\Delta_2},\tag{45}$$

which is a two-soliton solution of (34), where

$$\begin{aligned}\Delta_1=&\sigma\zeta_1^2\zeta_3^2\left(\zeta_3^2-\zeta_1^2\right)\left(\alpha_3^*\zeta_1^{*3}-\alpha_1^*\zeta_3^{*3}\right)\\&-\alpha_1^*\zeta_1^2\zeta_1^{*2}\left(\zeta_1^{*2}-\zeta_1^2\right)\left(\alpha_3\alpha_3^*\zeta_3^3-\sigma\zeta_3^{*3}\right)\\&+\alpha_3^*\zeta_1^2\zeta_3^{*2}\left(\zeta_3^{*2}-\zeta_1^2\right)\left(\alpha_3\alpha_1^*\zeta_3^3-\sigma\zeta_1^{*3}\right)\\&+\alpha_1^*\zeta_3^2\zeta_1^{*2}\left(\zeta_1^{*2}-\zeta_3^2\right)\left(\alpha_1\alpha_3^*\zeta_1^3-\sigma\zeta_3^{*3}\right)\\&-\alpha_3^*\zeta_3^2\zeta_3^{*2}\left(\zeta_3^{*2}-\zeta_3^2\right)\left(\alpha_1\alpha_1^*\zeta_1^3-\sigma\zeta_1^{*3}\right)\\&+\alpha_1^*\alpha_3^*\zeta_1^{*2}\zeta_3^{*2}\left(\zeta_3^{*2}-\zeta_1^{*2}\right)\left(\alpha_1\zeta_1^3-\alpha_3\zeta_3^3\right),\end{aligned}$$

$$\begin{aligned}\Delta_2=&|\zeta_1|^2|\zeta_3|^2\left[\left|\zeta_3^2-\zeta_1^2\right|^2\left(\zeta_1\zeta_3\right.\right.\\&\left.+|\alpha_1|^2|\alpha_3|^2\zeta_1^*\zeta_3^*\right)\\&-\sigma\left(\zeta_3^{*2}-\zeta_3^2\right)\left(\zeta_1^{*2}-\zeta_1^2\right)\left(\alpha_1^*\alpha_3|\zeta_1|^2\right.\\&\left.\left.+\alpha_1\alpha_3^*|\zeta_3|^2\right)\right.\\&\left.-\sigma\left|\zeta_3^{*2}-\zeta_1^2\right|^2\left(|\alpha_3|^2\zeta_1\zeta_3^*+|\alpha_1|^2\zeta_3\zeta_1^*\right)\right],\end{aligned}$$

$$\begin{aligned}&\alpha_j=-\gamma_j\exp\left(2i\zeta_j^2x+2i\eta_j^2t\right),\\&\eta_j=\sqrt{\alpha}\left(\zeta_j-\frac{\beta}{2\zeta_j}\right),\quad j=1,3.\end{aligned}$$

Plots of this solution are given in Fig. 2 if the parameters are chosen as $\alpha=1,\beta=-1,\sigma=-1,\zeta_1=0.48+0.3i,\gamma_1=0.05+0.15i,\zeta_3=0.064+0.37i,\gamma_3=0.98+0.99i$ (see Fig. 2).

Now we choose two seek solutions of (34) as follows:

$$u=c_0e^{i\sigma\gamma_0^2x},\qquad c_0=\frac{\beta+\sigma\gamma_0^2}{\beta\gamma_0},\tag{46}$$

$$\begin{aligned}&u=\frac{\omega_0}{\beta\gamma_0}e^{i\sigma(\gamma_0^2x+\delta_0t)},\\&\delta_0=\alpha\left[\left(\beta+\sigma\gamma_0^2\right)^2-\omega_0^2\right]\gamma_0^{-2},\end{aligned}\tag{47}$$

where γ_0 and ω_0 are constants.

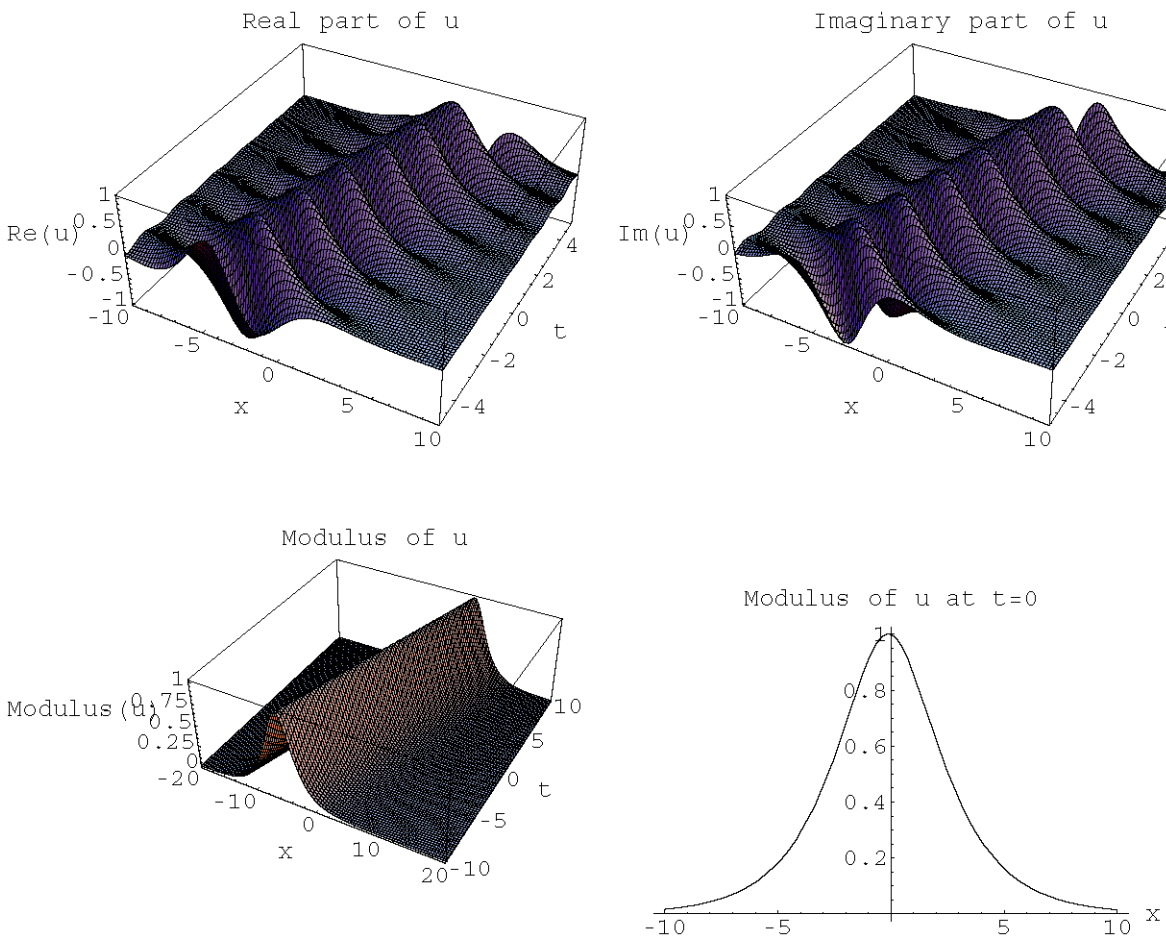


Fig. 1 One-soliton solution

Firstly, substituting the seek solution (46) of (34) into (3) and (4), their two basic solutions $\phi(\zeta)$ and $\psi(\zeta)$ are chosen as

$$\phi(\zeta)=\begin{pmatrix} \exp\left(\frac{i\sigma\gamma_0^2+\sqrt{\Delta_1(\zeta)}}{2}x+\frac{2\alpha\zeta^2\sigma\gamma_0^2+\alpha\beta^2}{4\zeta^2\sigma\gamma_0^2}\sqrt{\Delta_1(\zeta)}t\right) \\ \frac{i\sigma\gamma_0^2+\sqrt{\Delta_1(\zeta)}+2i\zeta^2}{2i\zeta\sigma\gamma_0^2c_0}\exp\left(\frac{-i\sigma\gamma_0^2+\sqrt{\Delta_1(\zeta)}}{2}x+\frac{2\alpha\zeta^2\sigma\gamma_0^2+\alpha\beta^2}{4\zeta^2\sigma\gamma_0^2}\sqrt{\Delta_1(\zeta)}t\right) \end{pmatrix},$$

$$\psi(\zeta)=\begin{pmatrix} \frac{-i\sigma\gamma_0^2+\sqrt{\Delta_1(\zeta)}-2i\zeta^2}{-2i\zeta\gamma_0^2c_0}\exp\left(\frac{i\sigma\gamma_0^2+\sqrt{\Delta_1(\zeta)}}{2}x+\frac{2\alpha\zeta^2\sigma\gamma_0^2+\alpha\beta^2}{4\zeta^2\sigma\gamma_0^2}\sqrt{\Delta_1(\zeta)}t\right) \\ \exp\left(\frac{-i\sigma\gamma_0^2+\sqrt{\Delta_1(\zeta)}}{2}x+\frac{2\alpha\zeta^2\sigma\gamma_0^2+\alpha\beta^2}{4\zeta^2\sigma\gamma_0^2}\sqrt{\Delta_1(\zeta)}t\right) \end{pmatrix}, \tag{48}$$

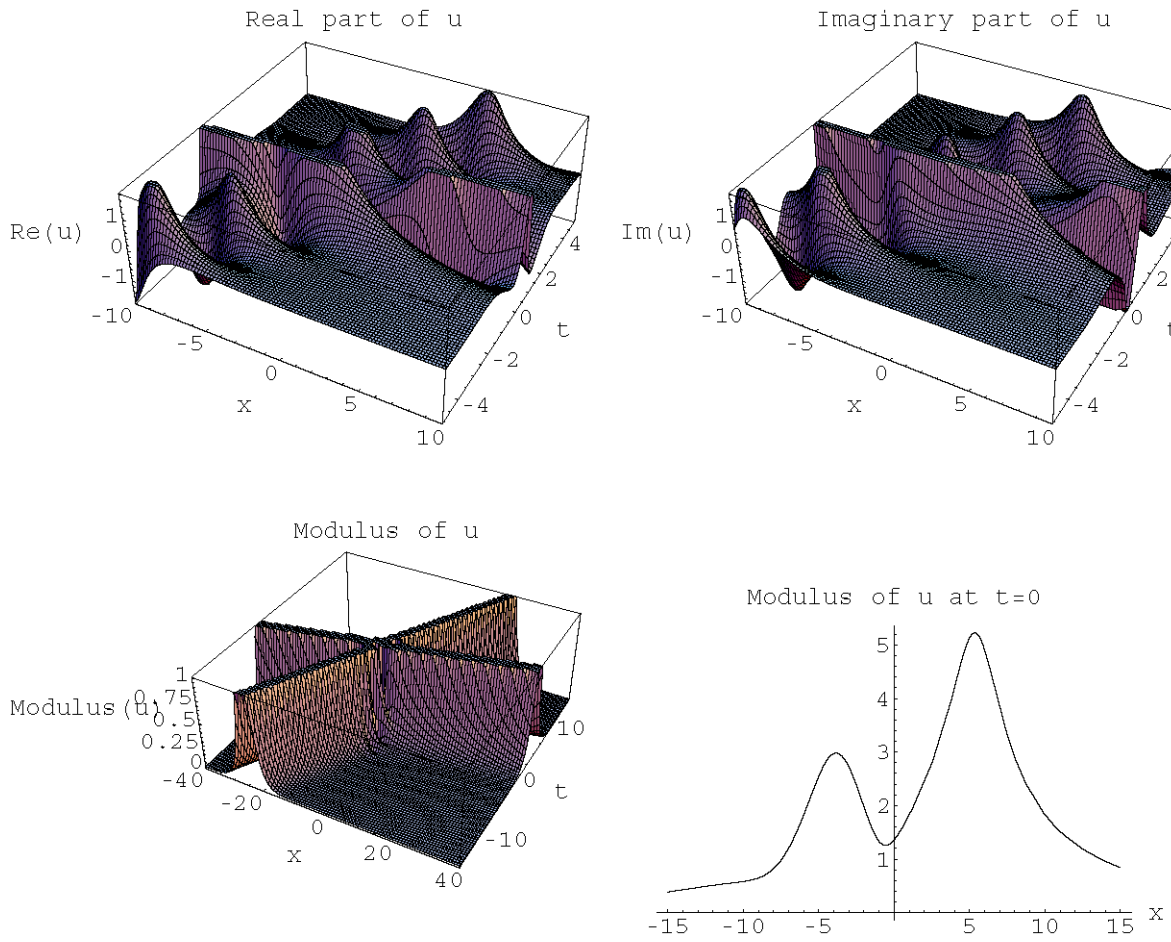


Fig. 2 Two-soliton solution

in the view of (35), where

$$\Delta_1(\zeta) = -\gamma_0^4 - 4\sigma\gamma_0^2\zeta^2 + 4\sigma\gamma_0^4\zeta^2 c_0^2 - 4\zeta^4.$$

Then (37) and (38) can be written as

$$\alpha_{2j-1} = \frac{i\sigma\gamma_0^2 + \sqrt{\Delta_1(\zeta_{2j-1})} + 2i\zeta_{2j-1}^2 - 2i\gamma_{2j-1}\zeta_{2j-1}\sigma\gamma_0^2 c_0}{2i\zeta_{2j-1}\sigma\gamma_0^2 c_0 - i\gamma_{2j-1}\gamma_0^2 + \sigma\gamma_{2j-1}\sqrt{\Delta_1(\zeta_{2j-1})} - 2i\sigma\gamma_{2j-1}\zeta_{2j-1}^2} e^{-i\sigma\gamma_0^2 x}, \tag{49}$$

$$\begin{aligned}
&\sum_{k=1}^{N}\left(A_k\zeta_{2j-1}^{2k} + \alpha_{2j-1}B_k\zeta_{2j-1}^{2k-1}\right) = -1,\\
&\sum_{k=1}^{N}\left(\alpha_{2j-1}^* A_k\zeta_{2j-1}^{*2k} + \sigma B_k\zeta_{2j-1}^{*2k-1}\right) = -\alpha_{2j-1}^*,\\
&j = 1,\ldots,N.
\end{aligned} \tag{50}$$

Using the DT, we obtain new exact solutions for the integrable generalization of the NLS equation (34):

$$\tilde{u} = u + B_1 = c_0 e^{i\sigma\gamma_0^2 x} + B_1, \qquad c_0 = \frac{\beta + \sigma\gamma_0^2}{\beta\gamma_0} \tag{51}$$

where B_1 is determined by (50).

(I) For $N = 1$, a direct calculation gives from (50) that

$$B_1=\frac{\alpha_1^*(\zeta_1^{*2}-\zeta_1^2)}{\zeta_1\zeta_1^*(\sigma\zeta_1-\alpha_1\alpha_1^*\zeta_1^*)}.$$

According to the DT (39), we get an explicit solution for the integrable generalization of the NLS equation (34)

$$\tilde{u}=c_0e^{i\sigma\gamma_0^2x}+\frac{\alpha_1^*(\zeta_1^{*2}-\zeta_1^2)}{\zeta_1\zeta_1^*(\sigma\zeta_1-\alpha_1\alpha_1^*\zeta_1^*)}, \tag{52}$$

with

$$\alpha_1=\frac{i\sigma\gamma_0^2+\sqrt{\Delta_1(\zeta_1)}+2i\zeta_1^2-2i\gamma_1\zeta_1\sigma\gamma_0^2c_0}{2i\zeta_1\sigma\gamma_0^2c_0-i\gamma_1\gamma_0^2+\sigma\gamma_1\sqrt{\Delta_1(\zeta_1)}-2i\sigma\gamma_1\zeta_1^2}e^{-i\sigma\gamma_0^2x},$$

$$\Delta_1(\zeta_1)=-\gamma_0^4-4\sigma\gamma_0^2\zeta_1^2+4\sigma\gamma_0^4\zeta_1^2c_0^2-4\zeta_1^4,$$
$$c_0=\frac{\beta+\sigma\gamma_0^2}{\beta\gamma_0}.$$

(II) For $N=2$, we obtain from linear algebraic system (50) that

$$\tilde{u}=u+B_1=c_0e^{i\sigma\gamma_0^2x}+\frac{\Delta_1}{\Delta_2}, \tag{53}$$

which is another explicit solution of (34), where

$$\begin{aligned}\Delta_1=&\sigma\zeta_1^2\zeta_3^2\big(\zeta_3^2-\zeta_1^2\big)\big(\alpha_3^*\zeta_1^{*3}-\alpha_1^*\zeta_3^{*3}\big)\\&-\alpha_1^*\zeta_1^2\zeta_1^{*2}\big(\zeta_1^{*2}-\zeta_1^2\big)\big(\alpha_3\alpha_3^*\zeta_3^3-\sigma\zeta_3^{*3}\big)\\&+\alpha_3^*\zeta_1^2\zeta_3^{*2}\big(\zeta_3^{*2}-\zeta_1^2\big)\big(\alpha_3\alpha_1^*\zeta_3^3-\sigma\zeta_1^{*3}\big)\\&+\alpha_1^*\zeta_3^2\zeta_1^{*2}\big(\zeta_1^{*2}-\zeta_3^2\big)\big(\alpha_1\alpha_3^*\zeta_1^3-\sigma\zeta_3^{*3}\big)\\&-\alpha_3^*\zeta_3^2\zeta_3^{*2}\big(\zeta_3^{*2}-\zeta_3^2\big)\big(\alpha_1\alpha_1^*\zeta_1^3-\sigma\zeta_1^{*3}\big)\\&+\alpha_1^*\alpha_3^*\zeta_1^{*2}\zeta_3^{*2}\big(\zeta_3^{*2}-\zeta_1^{*2}\big)\big(\alpha_1\zeta_1^3-\alpha_3\zeta_3^3\big),\end{aligned}$$

$$\begin{aligned}\Delta_2=&|\zeta_1|^2|\zeta_3|^2\big[\big|\zeta_3^2-\zeta_1^2\big|^2\big(\zeta_1\zeta_3\\&+|\alpha_1|^2|\alpha_3|^2\zeta_1^*\zeta_3^*\big)\\&-\sigma\big(\zeta_3^{*2}-\zeta_3^2\big)\big(\zeta_1^{*2}-\zeta_1^2\big)\big(\alpha_1^*\alpha_3|\zeta_1|^2\\&+\alpha_1\alpha_3^*|\zeta_3|^2\big)\\&-\sigma\big|\zeta_3^{*2}-\zeta_1^2\big|^2\big(|\alpha_3|^2\zeta_1\zeta_3^*+|\alpha_1|^2\zeta_3\zeta_1^*\big)\big],\end{aligned}$$

$$\alpha_j=\frac{i\sigma\gamma_0^2+\sqrt{\Delta_1(\zeta_j)}+2i\zeta_j^2-2i\gamma_j\zeta_j\sigma\gamma_0^2c_0}{2i\zeta_j\sigma\gamma_0^2c_0-i\gamma_j\gamma_0^2+\sigma\gamma_j\sqrt{\Delta_1(\zeta_j)}-2i\sigma\gamma_j\zeta_j^2}e^{-i\sigma\gamma_0^2x},$$

$$\Delta_1(\zeta_j)=-\gamma_0^4-4\sigma\gamma_0^2\zeta_j^2+4\sigma\gamma_0^4\zeta_j^2c_0^2-4\zeta_j^4,$$
$$j=1,3,\quad c_0=\frac{\beta+\sigma\gamma_0^2}{\beta\gamma_0}.$$

Secondly, substituting the seek solution (47) of (34) into (3) and (4), their two basic solutions $\phi(\zeta)$ and $\psi(\zeta)$ are taken as

$$\begin{aligned}\phi(\zeta)&=\begin{pmatrix}\exp\big(\frac{i\sigma\gamma_0^2+\sqrt{\Delta_2(\zeta)}}{2}x+\frac{2i\zeta^2\gamma_0^2\delta_0+(2\alpha\zeta^2\sigma\gamma_0^2+\alpha\beta^2)\sqrt{\Delta_2(\zeta)}}{4\zeta^2\sigma\gamma_0^2}t\big)\\ \frac{i\sigma\beta\gamma_0^2+\beta\sqrt{\Delta_2(\zeta)}+2i\beta\zeta^2}{2i\zeta\sigma\gamma_0\omega_0}\exp\big(\frac{-i\sigma\gamma_0^2+\sqrt{\Delta_2(\zeta)}}{2}x+\frac{-2i\zeta^2\gamma_0^2\delta_0+(2\alpha\zeta^2\sigma\gamma_0^2+\alpha\beta^2)\sqrt{\Delta_2(\zeta)}}{4\zeta^2\sigma\gamma_0^2}t\big)\end{pmatrix},\\ \psi(\zeta)&=\begin{pmatrix}\frac{-i\sigma\beta\gamma_0^2+\beta\sqrt{\Delta_2(\zeta)}-2i\beta\zeta^2}{-2i\zeta\gamma_0\omega_0}\exp\big(\frac{i\sigma\gamma_0^2+\sqrt{\Delta_2(\zeta)}}{2}x+\frac{2i\zeta^2\gamma_0^2\delta_0+(2\alpha\zeta^2\sigma\gamma_0^2+\alpha\beta^2)\sqrt{\Delta_2(\zeta)}}{4\zeta^2\sigma\gamma_0^2}t\big)\\ \exp\big(\frac{-i\sigma\gamma_0^2+\sqrt{\Delta_2(\zeta)}}{2}x+\frac{-2i\zeta^2\gamma_0^2\delta_0+(2\alpha\zeta^2\sigma\gamma_0^2+\alpha\beta^2)\sqrt{\Delta_2(\zeta)}}{4\zeta^2\sigma\gamma_0^2}t\big)\end{pmatrix}\end{aligned} \tag{54}$$

in the view of (35), where

$$\Delta_2(\zeta)=-\gamma_0^4-4\sigma\gamma_0^2\zeta^2+\frac{4\omega_0^2}{\beta^2}\sigma\gamma_0^2\zeta^2-4\zeta^4.$$

Then (37) and (38) can be written as

$$\alpha_{2j-1}=\frac{i\sigma\beta\gamma_0^2+\beta\sqrt{\Delta_2(\zeta_{2j-1})}+2i\beta\zeta_{2j-1}^2-2i\gamma_{2j-1}\zeta_{2j-1}\sigma\gamma_0\omega_0}{2i\zeta_{2j-1}\sigma\gamma_0\omega_0-i\beta\gamma_{2j-1}\gamma_0^2+\sigma\beta\gamma_{2j-1}\sqrt{\Delta_2(\zeta_{2j-1})}-2i\sigma\beta\gamma_{2j-1}\zeta_{2j-1}^2}e^{-i\sigma\gamma_0^2x-i\sigma\delta_0t}, \tag{55}$$

$$\sum_{k=1}^{N}\left(A_k\zeta_{2j-1}^{2k}+\alpha_{2j-1}B_k\zeta_{2j-1}^{2k-1}\right)=-1,$$
$$\sum_{k=1}^{N}\left(\alpha_{2j-1}^{*}A_k\zeta_{2j-1}^{*2k}+\sigma B_k\zeta_{2j-1}^{*2k-1}\right)=-\alpha_{2j-1}^{*},\quad j=1,\ldots,N. \tag{56}$$

Using the DT, we obtain new exact solutions for the integrable generalization of the NLS equation (34):

$$\tilde{u}=u+B_1=\frac{\omega_0}{\beta\gamma_0}e^{i\sigma(\gamma_0^2x+\delta_0t)}+B_1,$$
$$\delta_0=\alpha\left[\left(\beta+\sigma\gamma_0^2\right)^2-\omega_0^2\right]\gamma_0^{-2} \tag{57}$$

where B_1 is determined by (56).

(I) For $N=1$, a direct calculation gives from (56) that

$$B_1=\frac{\alpha_1^{*}(\zeta_1^{*2}-\zeta_1^2)}{\zeta_1\zeta_1^{*}(\sigma\zeta_1-\alpha_1\alpha_1^{*}\zeta_1^{*})}.$$

According to the DT (39), we get an explicit solution for the integrable generalization of the NLS equation (34)

$$\tilde{u}=\frac{\omega_0}{\beta\gamma_0}e^{i\sigma(\gamma_0^2x+\delta_0t)}+\frac{\alpha_1^{*}(\zeta_1^{*2}-\zeta_1^2)}{\zeta_1\zeta_1^{*}(\sigma\zeta_1-\alpha_1\alpha_1^{*}\zeta_1^{*})}, \tag{58}$$

with

$$\alpha_1=\frac{i\sigma\beta\gamma_0^2+\beta\sqrt{\Delta_2(\zeta_1)}+2i\beta\zeta_1^2-2i\gamma_1\zeta_1\sigma\gamma_0\omega_0}{2i\zeta_1\sigma\gamma_0\omega_0-i\beta\gamma_1\gamma_0^2+\sigma\beta\gamma_1\sqrt{\Delta_2(\zeta_1)}-2i\sigma\beta\gamma_1\zeta_1^2}e^{-i\sigma\gamma_0^2x-i\sigma\delta_0t},$$

$$\Delta_2(\zeta_1)=-\gamma_0^4-4\sigma\gamma_0^2\zeta_1^2+\frac{4\omega_0^2}{\beta^2}\sigma\gamma_0^2\zeta_1^2-4\zeta_1^4,$$
$$\delta_0=\alpha\left[\left(\beta+\sigma\gamma_0^2\right)^2-\omega_0^2\right]\gamma_0^{-2}.$$

(II) For $N=2$, we obtain from linear algebraic system (56) that

$$\tilde{u}=u+B_1=\frac{\omega_0}{\beta\gamma_0}e^{i\sigma(\gamma_0^2x+\delta_0t)}+\frac{\Delta_1}{\Delta_2}, \tag{59}$$

which is another explicit solution of (34), where

$$\begin{aligned}\Delta_1=&\sigma\zeta_1^2\zeta_3^2\left(\zeta_3^2-\zeta_1^2\right)\left(\alpha_3^{*}\zeta_1^{*3}-\alpha_1^{*}\zeta_3^{*3}\right)\\&-\alpha_1^{*}\zeta_1^2\zeta_1^{*2}\left(\zeta_1^{*2}-\zeta_1^2\right)\left(\alpha_3\alpha_3^{*}\zeta_3^3-\sigma\zeta_3^{*3}\right)\\&+\alpha_3^{*}\zeta_1^2\zeta_3^{*2}\left(\zeta_3^{*2}-\zeta_1^2\right)\left(\alpha_3\alpha_1^{*}\zeta_3^3-\sigma\zeta_1^{*3}\right)\\&+\alpha_1^{*}\zeta_3^2\zeta_1^{*2}\left(\zeta_1^{*2}-\zeta_3^2\right)\left(\alpha_1\alpha_3^{*}\zeta_1^3-\sigma\zeta_3^{*3}\right)\\&-\alpha_3^{*}\zeta_3^2\zeta_3^{*2}\left(\zeta_3^{*2}-\zeta_3^2\right)\left(\alpha_1\alpha_1^{*}\zeta_1^3-\sigma\zeta_1^{*3}\right)\\&+\alpha_1^{*}\alpha_3^{*}\zeta_1^{*2}\zeta_3^{*2}\left(\zeta_3^{*2}-\zeta_1^{*2}\right)\left(\alpha_1\zeta_1^3-\alpha_3\zeta_3^3\right),\end{aligned}$$

$$\begin{aligned}\Delta_2=&|\zeta_1|^2|\zeta_3|^2\left[\left|\zeta_3^2-\zeta_1^2\right|^2\left(\zeta_1\zeta_3\right.\right.\\&+|\alpha_1|^2|\alpha_3|^2\zeta_1^{*}\zeta_3^{*}\left.\right)\\&-\sigma\left(\zeta_3^{*2}-\zeta_3^2\right)\left(\zeta_1^{*2}-\zeta_1^2\right)\left(\alpha_1^{*}\alpha_3|\zeta_1|^2\right.\\&+\alpha_1\alpha_3^{*}|\zeta_3|^2\left.\right)\\&\left.-\sigma\left|\zeta_3^{*2}-\zeta_1^2\right|^2\left(|\alpha_3|^2\zeta_1\zeta_3^{*}+|\alpha_1|^2\zeta_3\zeta_1^{*}\right)\right],\end{aligned}$$

$$\alpha_j=\frac{i\sigma\beta\gamma_0^2+\beta\sqrt{\Delta_2(\zeta_j)}+2i\beta\zeta_j^2-2i\gamma_j\zeta_j\sigma\gamma_0\omega_0}{2i\zeta_j\sigma\gamma_0\omega_0-i\beta\gamma_j\gamma_0^2+\sigma\beta\gamma_j\sqrt{\Delta_2(\zeta_j)}-2i\sigma\beta\gamma_j\zeta_j^2}e^{-i\sigma\gamma_0^2x-i\sigma\delta_0t},$$

$$\Delta_2(\zeta_j)=-\gamma_0^4-4\sigma\gamma_0^2\zeta_j^2+\frac{4\omega_0^2}{\beta^2}\sigma\gamma_0^2\zeta_j^2-4\zeta_j^4,$$
$$j=1,3,\qquad \delta_0=\alpha\left[\left(\beta+\sigma\gamma_0^2\right)^2-\omega_0^2\right]\gamma_0^{-2}.$$

4 Conclusions

The integrable generalization of the NLS equation models the propagation of nonlinear light pulses in optical fibers when certain higher-order non-linear effects are taken into account. Therefore, the study of explicit solutions for this equation is interesting and significant. The present work is devoted to a study for the integrable generalization of the NLS equation by means of the DT. In the foregoing sections, we have given an approach to construct a DT for the integrable generalization of the coupled NLS equation based on a gauge transformation between the corresponding spectral problems. A DT for the integrable generalization of the NLS equation is obtained through the reduction technique. As applications of DT, two classes of new explicit solutions for the integrable generalization of the NLS equation are given explicitly.

Acknowledgements This work was supported by National Natural Science Foundation of China (Project no. 11171312).

References

1. Fokas, A.S.: On a class of physically important integrable equations. Physica D **87**, 145–150 (1995)
2. Lenells, L.: Exactly solvable model for nonlinear pulse propagation in optical fibers. Stud. Appl. Math. **123**, 215–232 (2009)
3. Lenells, L., Fokas, A.S.: On a novel integrable generalization of the nonlinear Schrödinger equation. Nonlinearity **22**, 11–27 (2009)
4. Lenells, L., Fokas, A.S.: An integrable generalization of the nonlinear Schrödinger equation on the half-line and solitons. Inverse Probl. **25**, 115006 (2009)

5. Lenells, L.: Dressing for a novel integrable generalization of the nonlinear Schrödinger equation. J. Nonlinear Sci. **20**, 709–722 (2010)
6. Matveev, V.B., Salle, M.A.: Darboux Transformations and Solitons. Springer, Berlin (1991)
7. Levi, D.: On a new Darboux transformation for the construction of exact solutions of the Schrödinger equation. Inverse Probl. **4**, 165–172 (1988)
8. Gu, C.H., Zhou, Z.X.: On Darboux transformations for soliton equations in high-dimensional spacetime. Lett. Math. Phys. **32**, 1–10 (1994)
9. Li, Y.S.: The reductions of the Darboux transformation and some solutions of the soliton equations. J. Phys. A **29**, 4187–4195 (1996)
10. Geng, X.G., Ham, H.W.: Darboux transformation and soliton solutions for generalized nonlinear Schrödinger equations. J. Phys. Soc. Jpn. **68**, 1508–1542 (1999)